

Notes: Mian, Sufi, and Verner (QJE, 2017)

Household Debt and Business Cycles Worldwide

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Country panel and debt variables	4
2.2	Macroeconomic outcomes and auxiliary data	4
2.3	Key descriptive facts	5
2.4	Timing and interpretation	5
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Recursive VAR: dynamic facts	6
3.2	Dynamic regression for leads of GDP growth	6
3.3	Jorda local projections	7
3.4	Main single-equation specification	7
3.5	Consumption and external-balance regressions	8
3.6	Proxy SVAR with mortgage-sovereign spreads	8
3.7	Cross-sectional IV episodes	9
3.8	Forecast-error specification	9
3.9	Global household debt cycle	10
4	Identification and interpretation (what makes the estimates credible)	10
4.1	What the main regressions identify	10
4.2	Credit demand versus credit supply	11
4.3	Rational expectations versus flawed expectations	11

4.4	Why macroeconomic frictions matter	11
4.5	Robustness and limitations	12
4.6	Relation to the literature	12
5	Tables and figures	12
5.1	Table I: summary statistics	12
5.2	Figure I and Table II: the dynamic boom-bust pattern	13
5.3	Figure II, Table III, and Table IV: main predictive result and robustness	14
5.4	Figure III: scatter plot of household debt and future growth	16
5.5	Table V: household debt finances consumption booms	16
5.6	Table VI and Figure IV: mortgage spreads and proxy SVAR	16
5.7	Table VII and Figure V: cross-sectional IV during the 2000s boom	17
5.8	Figure VI and Table VIII: forecast errors and expectations	18
5.9	Table IX and Table X: nonlinearities, exchange rates, and unemployment	19
5.10	Table XI: house prices	19
5.11	Table XII, Table XIII, and Figure VII: external adjustment and the global debt cycle	20
6	Short summary	21
7	Conclusion	22
7.1	Core conclusion	22
7.2	Economic intuition	22
7.3	Takeaway	22

1 Big picture

- **Question.** Do household debt booms predict business-cycle downturns across countries, and what mechanism best explains this boom-bust pattern?
- **Setting.** The paper studies an unbalanced annual panel of **30 mostly advanced countries** from **1960–2012**. The main variables are household debt to GDP, nonfinancial firm debt to GDP, GDP growth, unemployment, forecasts, interest spreads, trade flows, and house prices.
- **Core empirical fact.** A rise in household debt to GDP over a three-year period predicts lower GDP growth over the next three years. A one-standard-deviation increase in household debt to GDP, about **6.2 percentage points**, is associated with roughly a **2.1 percentage point** decline in subsequent three-year GDP growth.
- **Household debt versus firm debt.** Nonfinancial firm debt does not generate the same medium-run boom-bust cycle. Firm debt has a smaller and more immediate negative relation with output, while household debt is associated with a consumption boom followed by a delayed growth slowdown.
- **Main mechanism.** The evidence points toward **credit supply shocks**, rather than pure credit demand shocks. Household debt booms occur in low interest-spread environments, and credit-supply-induced debt booms predict lower future growth.
- **Expectations result.** Professional forecasters, including IMF and OECD forecasters, do not lower their forecasts after household debt booms. As a result, household debt booms predict **negative GDP forecast errors**. This suggests that expectations are systematically too optimistic near the end of debt booms.
- **Macroeconomic frictions.** The bust is stronger when adjustment is harder: the effect is non-linear, stronger under fixed exchange rate regimes, and accompanied by higher unemployment.
- **Global dimension.** There is a global household debt cycle. Countries whose household debt cycles are more correlated with the global cycle have less ability to cushion the bust through net exports.
- **Economic takeaway.** Household debt booms are not just benign consumption smoothing. They often reflect excessive credit supply, flawed expectations, and macroeconomic frictions that turn household deleveraging into weak growth and unemployment.

2 Data and measurement (key objects)

2.1 Country panel and debt variables

- **Sample.** The data are annual and cover 30 countries from 1960–2012, with more than 900 country-year observations before differencing.
- **Debt source.** The main debt variables come from the BIS “Long series on total credit to the nonfinancial sectors.”
- **Household debt.** Household debt to GDP is defined as

$$d_{it}^{HH} = \frac{D_{it}^{HH}}{Y_{it}},$$

where D_{it}^{HH} is outstanding credit to households and Y_{it} is GDP.

- **Nonfinancial firm debt.** Firm debt to GDP is defined analogously as

$$d_{it}^F = \frac{D_{it}^F}{Y_{it}}.$$

- **Main debt-change object.** The main explanatory variable is the three-year change in debt from the end of $t - 4$ to the end of $t - 1$:

$$\Delta_3 d_{it-1}^{HH} = d_{it-1}^{HH} - d_{it-4}^{HH}, \quad \Delta_3 d_{it-1}^F = d_{it-1}^F - d_{it-4}^F.$$

- **Why lag the debt boom?** Lagging the debt change by one year makes sure that the debt boom is already observable when future growth or forecast errors are measured.

2.2 Macroeconomic outcomes and auxiliary data

- **Output.** The key outcome is real GDP growth, usually measured as log real GDP growth from t to $t + 3$:

$$\Delta_3 y_{it+3} = y_{it+3} - y_{it}.$$

- **Labor-market outcome.** The paper also studies the change in unemployment from t to $t + 3$.
- **Consumption and trade.** The paper decomposes the boom into consumption, investment, net exports, current accounts, consumption imports, and consumption exports.
- **Forecast data.** The paper uses IMF World Economic Outlook forecasts and OECD Economic Outlook forecasts to study whether forecasters anticipate the post-debt-boom slowdown.
- **Interest spreads.** The key credit-supply proxy is the **mortgage-sovereign spread**, defined as the mortgage lending rate minus the 10-year government bond yield.

- **House prices.** The paper uses BIS real residential house price data to separate the role of house prices from the role of household debt.
- **Exchange-rate regimes.** The paper uses de facto exchange-rate classifications from Reinhart-Rogoff and Ilzetzi-Reinhart-Rogoff to examine whether monetary policy flexibility matters.

2.3 Key descriptive facts

- In the main sample, mean annual real GDP growth is about **2.90 percent**; mean three-year GDP growth is about **8.40 percent**.
- Mean three-year household debt growth is about **4.58 percentage points** of GDP, with standard deviation **6.24**.
- Mean three-year firm debt growth is about **3.89 percentage points** of GDP, with standard deviation **12.21**.
- Household debt growth is highly persistent: the serial correlation of three-year household debt growth is about **0.79**.
- The paper's main result is economically large because the household-debt coefficient is about -0.33 in the baseline regression. Multiplying this by the standard deviation of household debt growth, 6.24, gives roughly -2.1 percentage points of three-year GDP growth.

2.4 Timing and interpretation

- The paper studies **medium-run business-cycle dynamics**, not long-run financial development.
- The baseline timing is:
 - debt boom: $t - 4$ to $t - 1$;
 - future outcome: t to $t + 3$.
- This timing is motivated by the VAR: after a household debt shock, household debt keeps rising for about three years before reverting.
- The empirical design is predictive and macroeconomic. The paper is careful not to claim that every regression coefficient is a fully causal micro-level effect.

3 Models and empirical specifications (all equations + interpretation)

3.1 Recursive VAR: dynamic facts

The first empirical object is a country-panel VAR in log real GDP, firm debt to GDP, and household debt to GDP:

$$Y_{it} = \left(y_{it}, d_{it}^F, d_{it}^{HH} \right).$$

The structural VAR is written as

$$AY_{it} = a_i + \sum_{j=1}^p \alpha_j Y_{it-j} + \varepsilon_{it},$$

where a_i are country fixed effects and $p = 5$ lags.

The reduced-form VAR is

$$Y_{it} = c_i + \sum_{j=1}^p \delta_j Y_{it-j} + u_{it}.$$

- The authors use a Cholesky decomposition with ordering

$$y_{it}, \quad d_{it}^F, \quad d_{it}^{HH}.$$

- The VAR is **descriptive**, not the main causal identification strategy.
- The household debt impulse response shows a debt boom lasting about three to four years followed by reversal.
- GDP initially rises after a household-debt shock, but then declines when household debt begins to reverse.
- Firm debt shocks have a more immediate negative relation with GDP and do not display the same delayed boom-bust pattern.

Interpretation. The VAR motivates the paper’s main horizon: a household debt boom is measured over three years, and the growth slowdown is measured over the next three years.

3.2 Dynamic regression for leads of GDP growth

The paper then estimates the dynamic relation directly:

$$\Delta_3 y_{it+k} = \alpha_i + \beta^{HH} \Delta_3 d_{it-1}^{HH} + \beta^F \Delta_3 d_{it-1}^F + u_{it+k},$$

for $k = -1, 0, \dots, 5$.

- Holding the debt boom fixed from $t - 4$ to $t - 1$, the left-hand-side growth window is moved forward.
- Household debt is positively related to contemporaneous growth, but negatively related to future growth.
- In Table II, the coefficient on household debt is 0.176 for contemporaneous three-year growth but falls to about -0.337 for growth from t to $t + 3$ and about -0.410 for growth from $t + 1$ to $t + 4$.
- Firm debt has a shorter-run negative relation but weaker medium-run predictive power.

Interpretation. Household debt booms are associated with a temporary expansion followed by a delayed contraction. This is the main reduced-form boom-bust pattern.

3.3 Jorda local projections

To check the robustness of the impulse responses, the paper estimates local projections. In levels, for horizons $h = 1, \dots, 10$:

$$y_{it+h-1} = \alpha_i^h + X_{it-1}\Gamma^h + \sum_{j=1}^5 \beta_j^{HH,h} d_{it-j}^{HH} + \sum_{j=1}^5 \beta_j^{F,h} d_{it-j}^F + \sum_{j=1}^5 \delta_j^h y_{it-j} + \varepsilon_{it+h-1}^h.$$

The paper also estimates first-difference local projections:

$$\Delta_h y_{it+h-1} = \alpha_i^h + X_{it-1}\Gamma^h + \sum_{j=1}^5 \beta_j^{HH,h} \Delta d_{it-j}^{HH} + \sum_{j=1}^5 \beta_j^{F,h} \Delta d_{it-j}^F + \sum_{j=1}^5 \delta_j^h \Delta y_{it-j} + u_{it+h-1}^h.$$

- Local projections are used because they are flexible and allow simple robustness checks.
- The paper checks robustness to adding a time trend and excluding the Great Recession.
- Across specifications, the medium-run decline in growth after household debt booms remains.

Interpretation. The key result is not just a product of the recursive VAR or the Great Recession. The boom-bust shape survives multiple dynamic specifications.

3.4 Main single-equation specification

The main regression is

$$\Delta_3 y_{it+3} = \alpha_i + \beta^{HH} \Delta_3 d_{it-1}^{HH} + \beta^F \Delta_3 d_{it-1}^F + X_{it-1}\Gamma + \varepsilon_{it}.$$

- α_i are country fixed effects.

- X_{it-1} may include lagged GDP growth, government debt, net foreign liabilities, time trends, or year fixed effects.
- Standard errors are clustered by country and year.
- The baseline coefficient on household debt is about -0.337 ; the firm debt coefficient is about -0.041 .

Interpretation. A rise in household debt predicts lower future growth even after controlling for firm debt. The magnitude is large and robust.

3.5 Consumption and external-balance regressions

To understand what happens during the boom, the paper estimates regressions of contemporaneous changes in consumption, investment, and trade variables on one-year changes in debt:

$$\Delta Z_{it} = \alpha_i + \beta^{HH} \Delta d_{it}^{HH} + \beta^F \Delta d_{it}^F + u_{it}.$$

- A rise in household debt is associated with higher consumption to GDP.
- The rise includes nondurable consumption, durable consumption, and services.
- Investment to GDP is roughly flat.
- Net exports and the current account decline during the boom.
- The share of imports that are consumption goods rises.

Interpretation. Household debt booms finance consumption booms, not primarily investment booms. This matters because it is harder to interpret the boom as borrowing against higher future productivity.

3.6 Proxy SVAR with mortgage-sovereign spreads

The paper uses mortgage-sovereign spreads as an external instrument for credit-supply shocks to household debt.

Let the reduced-form VAR residual be

$$u_{it} = S\varepsilon_{it}.$$

The goal is to identify the structural household credit-supply shock ε_{it}^3 using an external instrument Z_{it} , where Z_{it} is based on the mortgage-sovereign spread.

The instrument validity conditions are:

$$E[Z_{it}\varepsilon_{it}^3] \neq 0, \quad E[Z_{it}\varepsilon_{it}^j] = 0 \quad \text{for } j \neq 3.$$

The first-stage idea is to regress the household-debt VAR residual on the spread instrument:

$$\hat{u}_{it}^{HH} = \pi Z_{it} + \eta_{it}.$$

- If low mortgage spreads predict higher household-debt residuals, this supports a credit-supply interpretation.
- The paper finds exactly this: low mortgage spreads are associated with positive household-debt innovations.
- Low mortgage spreads are not meaningfully related to firm-debt residuals, which helps the exclusion argument.

Interpretation. Credit-demand stories usually predict that more borrowing should come with higher spreads. The finding that debt booms occur when mortgage spreads are low points toward expanding credit supply.

3.7 Cross-sectional IV episodes

For the 2000s boom and Great Recession, the paper estimates cross-sectional IV regressions:

$$\begin{aligned}\Delta_{2002,2007}d_i^{HH} &= \alpha^f + \beta^f z_i + u_i^f, \\ \Delta_{2007,2010}y_i &= \alpha^s + \beta^s \widehat{\Delta d_i^{HH}} + u_i^s.\end{aligned}$$

Two instruments are used:

- real sovereign spread convergence for eurozone countries before the euro;
- the decline in the mortgage-sovereign spread during the early 2000s.

Interpretation. Countries where spreads fell more experienced larger household debt booms, and the debt-boom component predicted by spreads was associated with worse subsequent GDP performance.

3.8 Forecast-error specification

The forecast-error analysis asks whether professional forecasters incorporated household-debt booms into future growth forecasts. Let $y_{t+h|t}^f$ be a forecast made at time t , and define the forecast error as

$$e_{t+h|t} = y_{t+h} - y_{t+h|t}^f.$$

The paper estimates

$$e_{i,t+h|t} = \alpha_i + \beta^{HH} \Delta_3 d_{it-1}^{HH} + \beta^F \Delta_3 d_{it-1}^F + u_{it}.$$

- Household debt booms do not predict lower IMF or OECD growth forecasts.
- But they do predict negative forecast errors.
- This means forecasters are too optimistic after household debt booms.

Interpretation. The evidence is difficult to reconcile with fully rational expectations. It supports models in which credit markets or forecasters underweight the future downside risk of household debt booms.

3.9 Global household debt cycle

The paper defines a country's correlation with the global household debt cycle as

$$\rho_i^{Global} = \text{corr} \left(\Delta_3 d_{it}^{HH}, \frac{1}{N-1} \sum_{j \neq i} \Delta_3 d_{jt}^{HH} \right).$$

It also estimates country-level and global regressions such as

$$\Delta_3 y_{it+3} = \alpha_i + \beta^{HH} \Delta_3 d_{it-1}^{HH} + \beta^F \Delta_3 d_{it-1}^F + \beta^G \left(Global_{-i} \Delta_3 d_{t-1}^{HH} \right) + u_{it}.$$

At the global level, it estimates

$$\Delta_3 \bar{y}_{t+3} = \alpha + \beta \Delta_3 \bar{d}_{t-1}^{HH} + \gamma \Delta_3 \bar{d}_{t-1}^F + u_t.$$

Interpretation. If many countries have household debt hangovers at the same time, they cannot all export their way out of the slowdown. This helps explain why the global 2000s household debt boom predicted the global slowdown after 2007.

4 Identification and interpretation (what makes the estimates credible)

4.1 What the main regressions identify

- The main regressions identify a robust **predictive relation**: household debt booms forecast weaker medium-run growth.
- The paper is careful that the baseline regressions are not by themselves a clean natural experiment.
- The credibility comes from the combination of:
 - dynamic VAR and local-projection evidence;

- strong robustness across samples and specifications;
- comparison between household debt and firm debt;
- spread-based evidence pointing to credit supply;
- forecast-error evidence pointing to flawed expectations;
- exchange-rate and unemployment evidence pointing to macro frictions.

4.2 Credit demand versus credit supply

- **Credit-demand story.** Households borrow more because they expect higher future income or productivity. This predicts higher future growth and, if credit supply is fixed, higher spreads.
- **What the data show.** Household debt booms predict lower future growth and are associated with low mortgage spreads.
- **Credit-supply story.** Lenders become willing to lend more or lend more cheaply to the same borrowers. This predicts rising household debt, lower spreads, higher consumption, and later reversal.
- The spread evidence is therefore central: it helps distinguish between outward shifts in credit demand and outward shifts in credit supply.

4.3 Rational expectations versus flawed expectations

- In rational-expectations models, agents should understand that household debt booms predict future weakness if this relation is stable.
- The forecast evidence shows that professional forecasters do *not* lower growth forecasts after household debt booms.
- Instead, household debt booms predict negative forecast errors and later downward forecast revisions.
- This supports credit-market sentiment or neglected-risk models in which lenders and investors underweight downside risk during booms.

4.4 Why macroeconomic frictions matter

- Household debt booms are followed by weak demand, but weak demand need not reduce output if prices, wages, and monetary policy adjust freely.
- The paper finds the strongest effects under fixed exchange-rate regimes, where monetary policy and exchange-rate adjustment are constrained.

- It also finds higher subsequent unemployment, consistent with nominal rigidities and demand shortfalls.
- The effect is nonlinear: increases in household debt predict downturns, but declines in household debt do not symmetrically predict booms.

4.5 Robustness and limitations

- The result survives controls for lagged GDP growth, government debt, net foreign liabilities, house prices, time trends, and year fixed effects.
- It is present in developed and emerging economies, and it remains in the pre-2006 sample, though with smaller magnitude.
- The paper uses alternative standard errors, nonoverlapping samples, Arellano-Bond GMM, and alternative debt normalizations.
- Important limitations remain:
 - cross-country macro data cannot fully isolate every structural shock;
 - spread instruments may have other macro channels;
 - global-cycle evidence is based on a relatively short time series;
 - the paper studies medium-run business cycles, not the long-run benefits of financial development.

4.6 Relation to the literature

- The paper extends work on household leverage and the Great Recession by showing that the household debt-growth relation appears across countries and decades.
- It complements theories of deleveraging, demand externalities, nominal rigidities, and constrained monetary policy.
- It connects to credit-market sentiment and neglected-risk models by showing that debt booms predict forecast errors, not only financial returns or bank crashes.
- It also distinguishes household debt from firm debt, showing that the household sector plays a special role in medium-run business cycles.

5 Tables and figures

5.1 Table I: summary statistics

Read:

TABLE I
SUMMARY STATISTICS

	N	Mean	Median	Std. dev.	$\frac{\text{Std. dev.}}{\text{Std. dev. } (\Delta y)}$	Ser. corr.
Δy	695	2.90	3.08	2.98	1.00	0.29
$\Delta_3 y$	695	8.40	8.65	6.56	2.21	0.71
$\Delta d^{Private}$	695	3.11	2.52	6.96	2.34	0.39
$\Delta_3 d^{Private}$	695	8.52	7.28	16.04	5.39	0.74
Δd^{HH}	695	1.62	1.33	2.56	0.86	0.43
$\Delta_3 d^{HH}$	695	4.58	3.68	6.24	2.10	0.79
Δd^F	695	1.48	1.04	5.66	1.90	0.30
$\Delta_3 d^F$	695	3.89	3.11	12.21	4.10	0.69
$\Delta_3 d^{Gov}$	627	1.73	1.16	9.92	3.33	0.71
$\Delta_3 d^{Netforeign}$	636	-0.57	1.30	15.01	5.04	0.79
Δc	678	2.81	2.90	2.84	0.95	0.33
Δc^{dur}	469	4.91	5.35	9.27	3.12	0.21
Δc^{nondur}	469	1.53	1.47	2.53	0.85	0.26
$\Delta \frac{C}{Y}$	690	-0.06	-0.01	1.17	0.39	0.04
Δi	678	2.66	3.67	10.79	3.63	0.15
Δg	688	2.84	2.60	2.79	0.94	0.26
Δx	695	8.64	9.30	12.29	4.13	0.15
Δm	695	8.08	9.55	13.87	4.66	0.12
$\Delta \frac{NX}{Y}$	695	0.14	-0.01	2.11	0.71	0.03
$\Delta \frac{CA}{Y}$	648	0.08	-0.02	2.29	0.77	-0.01
Δs^{XC}	695	-0.15	-0.07	1.80	0.61	0.04
Δs^{MC}	695	0.16	0.15	1.67	0.56	0.00
$\Delta reer$	614	-0.03	0.59	6.75	2.27	0.05
Δu	669	0.08	-0.01	1.08	0.36	0.34
$\Delta_3 u$	662	0.19	-0.01	2.43	0.82	0.67
$\Delta_3 y_{t+3}^{WEO}$	484	9.41	8.60	3.76	1.26	0.50
$\Delta_3 (y_{t+3} - y_{t+3}^{WEO})$	484	-2.53	-1.79	5.35	1.80	0.54
$\Delta_3 \ln(PHousing)$	514	6.56	7.16	17.42	5.85	0.72
spr^{real}	622	0.43	0.40	2.11	0.71	0.42
spr^{MS}	517	1.15	0.99	1.52	0.51	0.45
spr^{corp}	460	0.76	0.65	1.03	0.35	0.42

Notes. Log changes and ratios are multiplied by 100 to report changes in percentages or percentage points. Δ and Δ_3 denote one-year and three-year changes, respectively. The variables y , $d^{Private}$, d^{HH} , d^F , d^{Gov} , $\Delta_3 d^{Netforeign}$, c , c^{dur} , c^{nondur} , $\frac{C}{Y}$, i , g , x , m , $\frac{NX}{Y}$, $\frac{CA}{Y}$, s^{XC} , s^{MC} , $reer$, u , y_{t+3}^{WEO} , $\ln(PHousing)$, spr^{real} , spr^{MS} and spr^{corp} denote log real GDP, private nonfinancial debt to GDP, household debt to GDP, nonfinancial firm debt to GDP, government debt to GDP, the change in net foreign liabilities (sum of current account deficits to GDP), log real consumption, log real durable consumption, log real nondurable consumption, consumption to GDP, log real investment, log real government consumption, log nominal exports, log nominal imports, net exports to GDP, current account to GDP, the share of consumption exports to total exports, the share of consumption imports to total imports, log real effective exchange rate, the unemployment rate, the IMF Fall World Economic Outlook time t forecast of growth from t to $t+3$, the log real house price index, the real 10-year government bond yield spread with respect to the United States, mortgage-sovereign spread, and the corporate-sovereign spread, respectively.

- The sample contains substantial variation in household and firm debt growth.
- Three-year household debt growth has mean **4.58** percentage points of GDP and standard deviation **6.24**.
- Three-year firm debt growth has a larger standard deviation, **12.21**, but later tables show it does not have the same medium-run predictive power for GDP.
- Household debt growth is persistent, with serial correlation about **0.79**; this is why debt booms naturally last several years.

5.2 Figure I and Table II: the dynamic boom-bust pattern

Read:

- Figure I shows that a household debt shock generates a three- to four-year debt boom followed by reversal.
- GDP rises temporarily after the household debt shock, but then falls when household debt reverses.
- Table II puts the dynamic pattern into regressions. The coefficient on household debt is positive for contemporaneous growth but turns negative for future growth.

TABLE II

CREDIT EXPANSION AND CONTEMPORANEOUS AND FUTURE THREE-YEAR GDP GROWTH

	Dependent variable: Δy_{t+k} , $k = -1, 0, \dots, 5$						
	Δy_{t-1} (1)	Δy_t (2)	Δy_{t+1} (3)	Δy_{t+2} (4)	Δy_{t+3} (5)	Δy_{t+4} (6)	Δy_{t+5} (7)
Δy_{t-1}^{HH}	0.176*	0.121	-0.0136	-0.178**	-0.337**	-0.410**	-0.405**
	(0.0793)	(0.0810)	(0.0680)	(0.0629)	(0.0779)	(0.0905)	(0.102)
Δy_{t-1}^F	-0.0430	-0.140*	-0.159**	-0.108**	-0.0411	0.0327	0.0876*
	(0.0556)	(0.0550)	(0.0437)	(0.0362)	(0.0349)	(0.0395)	(0.0373)
Country fixed effects	✓	✓	✓	✓	✓	✓	✓
Test for equality of β_{HH} and β_F , p-value	0.465	0.184	0.0905	0.358	0.017	0.002	0.002
R^2	0.0256	0.0631	0.0999	0.103	0.128	0.138	0.128
Observations	815	785	755	725	695	665	635

Notes: This table presents results from estimating the following specification $\Delta y_{t+k} = \alpha_0 + \beta_{HH} \Delta y_{t-1}^{HH} + \beta_F \Delta y_{t-1}^F + \eta_{t+k}$ for $k = -1, \dots, 5$. Each column gradually leads the left-hand-side variable by one year. Reported R^2 values are from within-country variation. Standard errors in parentheses are dually clustered on country and year. *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

- The Table II coefficient on household debt is about -0.337 for growth from t to $t+3$ and about -0.410 for growth from $t+1$ to $t+4$.
- Firm debt does not show the same delayed boom-bust pattern.

5.3 Figure II, Table III, and Table IV: main predictive result and robustness

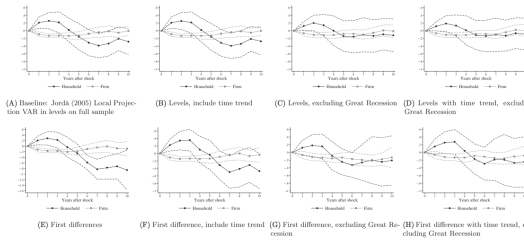


FIGURE II
Robustness with Jordà (2005) Local Projection Impulse Responses

This figure presents impulse responses from Jordà (2005) local projections estimated in levels and first differences. The specification in levels is: $y_{t+h-1} = \alpha_0 + \lambda_{0-1} \Gamma + \sum_{j=1}^5 \beta_{HH} \Delta y_{t-j}^{HH} + \sum_{j=1}^5 \beta_F \Delta y_{t-j}^F + \sum_{j=1}^5 \beta_{NG} \Delta y_{t-j}^{NG} + \epsilon_{t+h-1}$, for horizons $h = 1, \dots, 10$. To be consistent with the VAR model, Δy_t^F is nominal debt in sector $s = HH, F$ in year t scaled by nominal GDP in $t-1$. The model in first differences is $\Delta y_{t+h-1} = \alpha_0 + \lambda_{0-1} \Gamma + \sum_{j=1}^5 \beta_{HH} \Delta y_{t-j}^{HH} + \sum_{j=1}^5 \beta_F \Delta y_{t-j}^F + \sum_{j=1}^5 \beta_{NG} \Delta y_{t-j}^{NG} + \epsilon_{t+h-1}$, for horizons $h = 1, \dots, 10$, where here Δy_t^F refers to the debt to GDP ratio in sector $s = HH, F$. Models that exclude the Great Recession use data up to 2006. Dashed lines represent 95% confidence intervals computed using standard errors dually clustered on country and year.

Read:

- Figure II shows that the medium-run GDP decline after household debt booms remains under local projections, time trends, and exclusion of the Great Recession.
- Table III is the main regression table. The baseline household debt coefficient is about -0.337 , while the firm debt coefficient is close to zero once both are included.
- A one-standard-deviation increase in household debt growth predicts about **2.1 percentage points** lower three-year GDP growth.
- Controlling for lagged GDP growth, government debt, and net foreign liabilities does not remove the household-debt effect.
- Table IV shows robustness across nonoverlapping samples, Arellano-Bond GMM, no country fixed effects, moving-block-bootstrap standard errors, time trends, year fixed effects, alternative normalization, developed countries, emerging countries, and pre-Great-Recession samples.

TABLE III

HOUSEHOLD DEBT EXPANSION PREDICTS LOWER SUBSEQUENT GROWTH

	Dependent variable: Δy_{t+3}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta y_{t-1}^{HH, \text{coun}}$	-0.119** (0.0313)							
Δy_{t-1}^{HH}		-0.366** (0.0772)						
Δy_{t-1}^F			-0.0978* (0.0391)	-0.337** (0.0779)	-0.333** (0.0771)	-0.340** (0.0868)	-0.325** (0.0830)	-0.192* (0.0909)
Δy_{t-1}^{NG}				-0.0411 (0.0349)	-0.0464 (0.0354)	-0.0235 (0.0437)	-0.0519 (0.0395)	-0.0498 (0.0380)
$\Delta y_{t-1}^{NG, \text{coun}}$						0.0534 (0.0430)		
$\Delta y_{t-1}^{NG, \text{coun}}$							0.00793 (0.0523)	
$\mathbf{1}(\Delta y_{t-1}^{NG, \text{coun}} > 0)$								0.736 (1.005)
$\Delta y_{t-1}^{HH} + \mathbf{1}(\Delta y_{t-1}^{NG, \text{coun}} > 0)$								-0.235* (0.140)
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Distributed lag in Δy								
Test for equality of β_{HH} and β_F , p-value				0.002	0.003	0.003	0.007	
R^2	0.0869	0.123	0.0364	0.128	0.131	0.126	0.168	0.181
Observations	695	695	695	695	695	627	636	636

Notes: This table reports regressions of real GDP growth from t to $t+3$ on the change in total private, household, and nonfinancial firm debt to GDP from the end of $t-4$ to the end of $t-1$. Columns (3)-(8) control for three lags of GDP growth over the same period as the change in debt to GDP, Δy_{t-1} , Δy_{t-2} , and Δy_{t-3} . Column (6) includes the increase in government debt to GDP over the same period, and column (7) controls for the change in net foreign debt, calculated as the sum of current account deficits to GDP over the same three-year period. Column (8) interacts the increase in household debt with a dummy for whether the cumulated current account over the same period is negative. All specifications include country fixed effects. Reported R^2 values are from within-country variation. Standard errors in parentheses are dually clustered on country and year. *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

TABLE IV
HOUSEHOLD DEBT EXPANSION PREDICTS LOWER GROWTH: ROBUSTNESS AND
SUBSAMPLES

Panel A: Robustness to alternative specifications							
	Dependent variable: Δy_{it+3}						
	OLS (1)	AB-GMM (2)	OLS (3)	MBB SE (4)	OLS (5)	OLS (6)	OLS (7)
$\Delta_3 d_{it-1}^{HH}$	-0.32** (0.081)	-0.32** (0.11)	-0.31** (0.087)	-0.33** (0.083)	-0.23** (0.075)	-0.21** (0.063)	
$\Delta_3 d_{it-1}^F$	-0.057 (0.044)	-0.069 (0.071)	-0.063 (0.045)	-0.046 (0.031)	-0.042 (0.033)	-0.037 (0.033)	
Trend					-0.23** (0.063)		
$\Delta_3 d_{it-1}^{HH}$, alt. norm.							-0.30** (0.070)
$\Delta_3 d_{it-1}^F$, alt. norm.							0.020 (0.034)
Country fixed effects	✓		✓	✓	✓	✓	✓
Distributed lag in Δy	✓	✓	✓	✓	✓	✓	✓
Year fixed effects						✓	✓
Sample	Nonoverl.	Nonoverl.	Full	Full	Full	Full	Full
Test for equality of β_{HH} and β_F , p-value	.007	.0004	.035	.0043	.031	.012	.0002
R^2	0.15		0.18	0.13	0.26	0.50	0.15
Observations	232	203	695	695	695	695	695
Panel B: Subsamples							
	(1)	(2)	(3)	(4)	(5)	(6)	
$\Delta_3 d_{it-1}^{HH}$	-0.37** (0.097)	-0.24** (0.070)	-0.29** (0.10)	-0.22** (0.054)	-0.19** (0.072)	-0.16** (0.057)	
$\Delta_3 d_{it-1}^F$	-0.027 (0.040)	-0.069 (0.063)	-0.038 (0.062)	-0.053 (0.039)	-0.055 (0.038)	-0.061 (0.038)	
Trend					-0.16* (0.066)		
Country fixed effects	✓	✓	✓	✓	✓	✓	
Distributed lag in Δy	✓	✓	✓	✓	✓	✓	
Year fixed effects						✓	
Sample	Developed	Emerging	Pre-1995	Pre-2006	Pre-2006	Pre-2006	
Test for equality of β_{HH} and β_F , p-value	.005	.082	.009	.021	.116	.17	
R^2	0.16	0.11	0.13	0.088	0.16	0.37	
Observations	529	166	259	517	517	517	

Notes. Panel A presents a variety of robustness tests of the main specification in Table III. Column (1) uses the nonoverlapping sample that only includes every third year. This specification controls for $\Delta_3 y_{it-1}$. Column (2) uses the Arellano-Bond GMM estimator for the equation in differences on the same nonoverlapping sample. We instrument for $\Delta_3 d_{it-1}^{HH}$ and $\Delta_3 d_{it-1}^F$ with a double lag, $\Delta_3 d_{it-4}^{HH}$ and $\Delta_3 d_{it-4}^F$. Column (3) omits country fixed effects from the main specification estimated using the full sample. Column (4) computes standard errors using the panel moving blocks bootstrap (Gonçalves 2011) with a block length of $l = 3$. Column (5) includes a time trend, and column (6) includes country fixed effects. Column (7) replaces the three-year change in debt to GDP with the change in debt normalized by initial $(t - 4)$ GDP. Panel B reports estimates on various subsamples. Columns (1) and (2) present separate estimates for developed and emerging economies. Emerging market economies are the Czech Republic, Hong Kong, Hungary, Indonesia, Korea, Mexico, Poland, Singapore, Thailand, and Turkey. Developed economies are the remaining countries. Column (3) uses data up to 1995, and columns (4)-(6) use data up to 2006. Column (5) includes a time trend, and column (6) controls for year fixed effects. Standard errors in all columns (except Panel A, column (4)) are dually clustered on country and year. Reported R^2 values in regressions including country fixed effects are from within-country variation. +, *, ** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

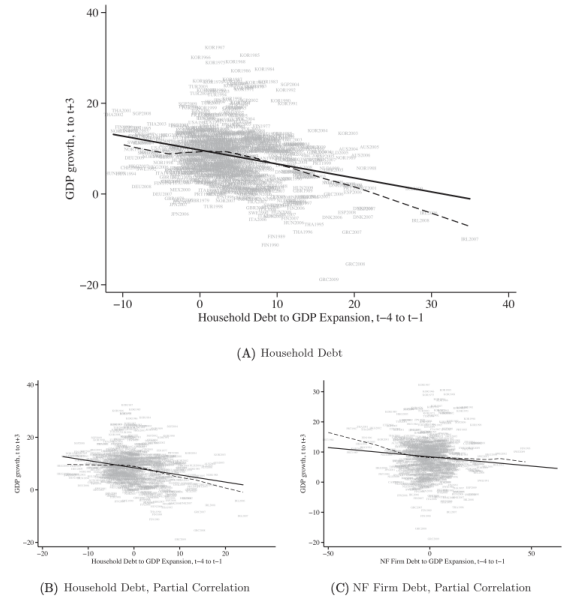


FIGURE III
Household Debt to GDP Expansion and Growth

This figure plots the relationship between GDP growth from t to $t + 3$ and the expansion in household and firm debt to GDP from $t - 4$ to $t - 1$. Each point refers to year t . The dashed line is the nonparametric plot of GDP growth from t to $t + 3$ against the increase in household or firm debt to GDP from $t - 4$ to $t - 1$. In Panel B household debt is partialled out with the expansion in nonfinancial firm debt to GDP, while in Panel C nonfinancial firm debt is partialled out with the expansion in household debt to GDP.

	$\Delta_1 \hat{Y}_H$ (1)	$\Delta_1 \hat{C}_{nondur}^H$ (2)	$\Delta_1 \hat{C}_{dur}^H$ (3)	$\Delta_1 \hat{C}_{inv}^H$ (4)	$\Delta_1 \hat{Y}_H$ (5)	$\Delta_1 \hat{N}_H$ (6)	$\Delta_1 \hat{C}_H^A$ (7)	$\Delta_1 \hat{M}_H^{CC}$ (8)	$\Delta_1 \hat{I}_H^{CC}$ (9)
$\Delta_1 d_H^{HH}$	0.120** (0.0462)	0.0432** (0.0152)	0.0333** (0.00701)	0.0709** (0.0230)	0.0174 (0.0756)	-0.173** (0.0582)	-0.185* (0.0813)	0.152** (0.0500)	0.0371 (0.0365)
$\Delta_1 d_H^F$	0.0249* (0.0146)	0.0200* (0.00781)	-0.0161** (0.00238)	0.0293** (0.00923)	-0.0194 (0.0264)	-0.0167 (0.0247)	-0.0125 (0.0207)	-0.0261 (0.0204)	-0.0400* (0.0197)
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
R^2	0.0825	0.0802	0.0647	0.138	0.00216	0.0408	0.0374	0.0417	0.0129
Observations	690	466	466	466	688	695	648	695	695

Notes. This table shows the contemporaneous correlation between the change in household and firm debt to GDP and the change in total consumption to GDP, nondurable consumption to GDP, durable consumption to GDP, services consumption to GDP, investment to GDP, net exports to GDP, current account to GDP, the share of consumption imports in total imports, and the share of consumption exports in total exports. All specifications include country fixed effects. Reported R^2 values are from within-country variation. Standard errors in parentheses are dually clustered on country and year. *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

5.4 Figure III: scatter plot of household debt and future growth

Read:

- Figure III shows the negative relation visually.
- Ireland and Greece during the Great Recession matter, but they are not the whole story.
- Other episodes, including Finland and Thailand, also help identify the relation.
- The nonparametric plot suggests nonlinearity: large positive household debt expansions are especially associated with lower future growth.

5.5 Table V: household debt finances consumption booms

Read:

- A rise in household debt is associated with higher total consumption to GDP.
- The rise in consumption is broad: nondurables, durables, and services all increase.
- Investment to GDP does not rise meaningfully.
- Net exports and the current account deteriorate during the boom.
- The share of imports devoted to consumption goods rises, consistent with debt-financed consumption rather than productivity-driven investment.

5.6 Table VI and Figure IV: mortgage spreads and proxy SVAR

Read:

- Table VI is the main first-stage evidence for the credit-supply interpretation.
- The mortgage-sovereign spread residual has coefficient -0.341 on the household-debt VAR residual; the low-spread indicator has coefficient 0.689 .
- The first-stage F -statistics are about **11.37** and **9.83**.
- The same spread variables are essentially unrelated to firm-debt residuals.

TABLE VI
PROXY SVAR FIRST-STAGE REGRESSIONS

	Residual from VAR household debt equation		Residual from VAR firm debt equation	
	$\hat{u}_{it}^{d^{HH}}$ (1)	$\hat{u}_{it}^{d^{HH}}$ (2)	$\hat{u}_{it}^{d^F}$ (3)	$\hat{u}_{it}^{d^F}$ (4)
MS spread, residual	-0.341** (0.101)		-0.0182 (0.267)	
Low MS spread indicator, residual		0.689** (0.220)		0.0347 (0.588)
F-statistic	11.372	9.834	0.005	0.003
R ²	0.024	0.021	0	0
Observations	580	580	580	580

Notes. This table shows regressions of the reduced-form VAR residuals on the mortgage-sovereign spread instruments. The MS spread residual is the residual from a regression of the MS spread on the VAR independent variables (including country fixed effects). The low MS spread indicator is the residual of a dummy variable that equals 1 if the standardized spread is below the sample median. Standard errors in parentheses account for contemporaneous correlation in residuals across countries, consistent with the bootstrap resampling procedure used to construct confidence intervals on the proxy SVAR impulse responses in Figure IV. +, *, ** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

- Figure IV shows that a low-mortgage-spread-induced household debt shock raises output briefly but is followed by a sharp medium-run decline.

5.7 Table VII and Figure V: cross-sectional IV during the 2000s boom

TABLE VII
DECLINING SPREADS, CREDIT EXPANSION, AND OUTPUT: SOVEREIGN SPREAD CONVERGENCE IN THE EUROZONE AND FALLING MORTGAGE SPREADS DURING THE 2000s CREDIT BOOM

	Eurozone case and sovereign spread over U.S.				2000s boom and mortgage-sovereign spread			
	Δ_{07-10Y} (1)	Δ_{02-07Y} (2)	Δ_{07-10Y} (3)	Δ_{07-10Y} (4)	Δ_{07-10Y} (5)	Δ_{02-07Y} (6)	Δ_{07-10Y} (7)	Δ_{07-10Y} (8)
$\Delta_{00-05Y} \text{pr}^{rel}$		-11.66** (3.428)						
$\Delta_{02-07Y} d^{HH}$	-0.170** (0.0404)		-0.222** (0.0478)	-0.218* (0.107)	-0.180 (0.118)		-0.296* (0.144)	-0.347 (0.306)
$\Delta_{02-07Y} d^F$				0.0326 (0.0833)				0.0975 (0.179)
Δ_{02-07Y}				-12.76 (14.36)				0.418** (0.103)
$\Delta_{00-05Y} \text{pr}^{HS}$						-10.28** (2.889)		
Equation	OLS	FS	IV	IV	OLS	FS	IV	IV
First stage F-statistic		11.6				12.609		
R ²	0.530	0.526	0.480	0.537	0.164	0.398	0.0952	0.362
Observations	12	12	12	12	21	21	21	21

Notes. This table reports instrumental variables regressions of GDP growth from 2007 to 2010 on the expansion in household debt to GDP from 2002 to 2007. Column (1) shows the OLS estimate for the eurozone countries and Denmark (in ERM II). Columns (2)-(4) use the change in the real sovereign spread (nominal spread minus inflation difference) with respect to the United States during 1996-1999 as an instrument for the 2002-2007 expansion in household or firm debt to GDP. Column (5) shows the OLS estimate for the broader sample of countries for which the mortgage-sovereign spread variable is nonmissing. Columns (6)-(8) use the change in the mortgage spread over the 10-year government bond yield during 2000-2004 as an instrument for the increase in household or firm debt to GDP from 2002 to 2007. The regressions in columns (6)-(10) exclude Hungary, as Hungary's decline in the mortgage-sovereign spread is a large outlier in this period. Standard errors in parentheses are robust to heteroskedasticity. +, *, ** indicate significance at the 0.1, 0.05, 0.01 levels respectively.

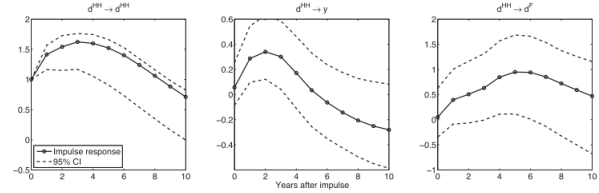
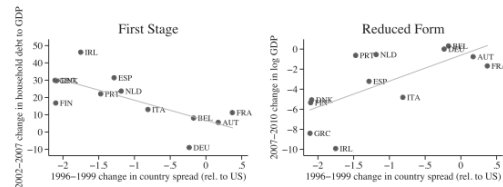


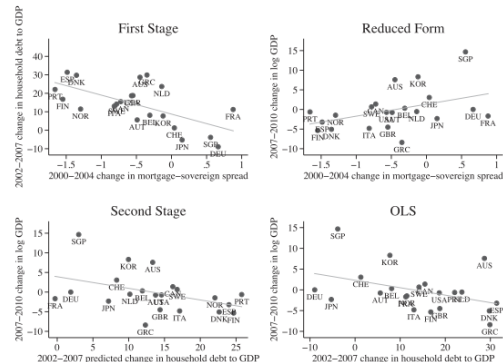
FIGURE IV

Impulse Responses to a Household Debt Shock Identified with the Mortgage Lending Spread in a Proxy SVAR

This figure shows impulse responses to a household debt shock identified using an indicator variable for whether the standardized mortgage spread is below the median as an external instrument in a proxy SVAR. The reduced-form VAR coefficient estimates are corrected for Nickell bias using an iterative bootstrap procedure. Dashed lines represent 95% confidence intervals that account for contemporaneous cross-country residual correlation and are computed by resampling cross-sections of residuals using the wild bootstrap.



(A) Eurozone Case and Sovereign Spread over U.S. 10-Year Treasury



(B) 2000s Mortgage Credit Boom and Mortgage-Sovereign Spread

FIGURE V

Declining Spreads, Credit Growth, and Output Growth

Read:

- In the eurozone, countries with larger pre-euro real sovereign spread declines experienced larger household debt booms from 2002–2007.
- The eurozone first-stage coefficient is about -11.66 , with first-stage F about **11.6**.
- The IV estimate says the spread-induced household debt boom predicted weaker GDP growth from 2007–2010; the IV coefficient is about -0.222 .
- In the broader 2000s sample, mortgage-spread declines also predict household debt booms. The first-stage coefficient is about -10.28 , with first-stage F about **12.67**.
- The corresponding IV effect of household debt on 2007–2010 growth is about -0.296 .

5.8 Figure VI and Table VIII: forecast errors and expectations

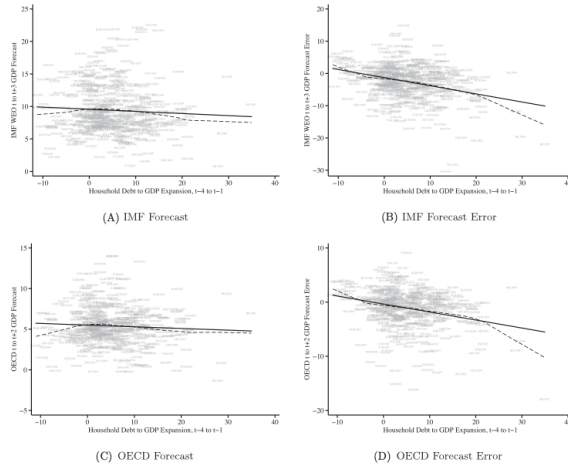


FIGURE VI

Household Debt Expansion Predicts Negative GDP Growth Forecast Errors

Panels A and B plot the three-year GDP forecasts and forecast errors from the fall issue of the IMF World Economic Outlook against the change in household debt to GDP from $t - 4$ to $t - 1$. The IMF WEO sample for the three-year-ahead forecasts includes years 1990–2012. Panels C and D plot the two-year GDP forecasts and forecast errors from the fall OECD Economic Outlook against the change in household debt to GDP from $t - 4$ to $t - 1$. The OECD sample includes years 1987–2012. We exclude Hong Kong, Indonesia, Singapore, and Thailand from the OECD sample because of gaps in the forecast series. Each point refers to year t .

TABLE VIII.
RISE IN HOUSEHOLD DEBT PREDICTS OVEROPTIMISTIC IMF AND OECD GROWTH FORECASTS

	Growth forecast		Forecast error					Forecast error sample up to 2006	
	Δy_{t-1}^{IMF} (1)	Δy_{t-1}^{OECD} (2)	e_{t-1}^{IMF} (3)	e_{t-1}^{IMF} (4)	e_{t-1}^{IMF} (5)	e_{t-1}^{OECD} (6)	e_{t-1}^{OECD} (7)	e_{t-1}^{IMF} (8)	e_{t-1}^{OECD} (9)
Δy_{t-1}^{IMF}	0.0016 (0.023)	0.0013 (0.028)	-0.060** (0.020)	-0.17** (0.057)	-0.31** (0.091)	-0.070** (0.023)	-0.17* (0.071)	-0.035* (0.021)	-0.042** (0.015)
Δy_{t-1}^{OECD}	-0.029 (0.021)	-0.041* (0.017)	-0.019 (0.027)	-0.026 (0.045)	-0.031 (0.061)	-0.013 (0.015)	-0.0084 (0.029)	-0.029 (0.020)	-0.020* (0.0080)
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Sample	Full	Full	Full	Full	Full	Full	Full	Pre-2006	Pre-2006
Test for equality of β_{IMF} and β_{OECD}	0.367	0.227	0.311	0.089	0.02	0.053	0.07	0.863	0.29
R^2	0.034	0.064	0.026	0.063	0.13	0.040	0.073	0.026	0.027
Observations	484	471	590	484	484	594	471	469	490

Notes: This table reports regression estimates of GDP growth forecasts and forecast errors on the change in household and nonfinancial firm debt to GDP from $t - 4$ to $t - 1$. The forecasts are from the fall issue of the IMF World Economic Outlook and the OECD Economic Outlook. Δy_{t-1}^{IMF} is the forecasted change in log GDP from $t - 4$ to $t - 1$. The forecasts are from the fall issue of the IMF World Economic Outlook and the OECD Economic Outlook. Δy_{t-1}^{OECD} is the forecasted change in log GDP from $t - 4$ to $t - 1$. The forecasts are from the fall issue of the IMF World Economic Outlook and the OECD Economic Outlook. e_{t-1}^{IMF} is the realized forecast error. The IMF and OECD forecast errors are constructed using the realized log GDP change reported in the IMF's Historical WEO Forecasts Database and the OECD Economic Outlook reports, respectively. The World Economic Outlook forecast sample includes all 70 countries in the sample and covers the years 1990–2012, with one-year-ahead forecasts extending back to 1972 for the G7. One- and two-year-ahead OECD Economic Outlook forecasts are for years 1978–2012 and 1987–2012, respectively. We exclude Hong Kong, Indonesia, Singapore, and Thailand from the OECD sample because of gaps in the forecast series. The regressions in columns (8) and (9) are estimated using data to 2006. Reported R^2 values are from within-country variation. Standard errors in parentheses are dually clustered on country and year. *, **, *** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

Read:

- IMF and OECD forecasts do not systematically fall after household debt booms.
- But household debt booms predict negative forecast errors.
- In Table VIII, the IMF forecast-error coefficient is about -0.060 for one-year-ahead errors, -0.17 for two-year-ahead errors, and -0.31 for three-year-ahead errors.
- OECD forecast errors show a similar pattern.
- This evidence supports the idea that the risks of household debt booms are underestimated in real time.

5.9 Table IX and Table X: nonlinearities, exchange rates, and unemployment

TABLE IX
NONLINEARITY AND HETEROGENEITY ACROSS EXCHANGE RATE REGIMES

	Nonlinearity	Fixed	Intermediate	Freely floating
	$\Delta_3 y_{it+3}$ (1)	$\Delta_3 y_{it+3}$ (2)	$\Delta_3 y_{it+3}$ (3)	$\Delta_3 y_{it+3}$ (4)
$\Delta_3 d_{it-1}^{HH} * 1(\Delta_3 d_{it-1}^{HH} > 0)$	-0.436** (0.106)			
$\Delta_3 d_{it-1}^{HH} * 1(\Delta_3 d_{it-1}^{HH} \leq 0)$	0.0655 (0.156)			
$\Delta_3 d_{it-1}^F * 1(\Delta_3 d_{it-1}^F > 0)$	-0.0537 (0.0367)			
$\Delta_3 d_{it-1}^F * 1(\Delta_3 d_{it-1}^F \leq 0)$	-0.0396 (0.0631)			
$\Delta_3 d_{it-1}^{HH}$		-0.534** (0.128)	-0.311** (0.0716)	-0.0673 (0.129)
$\Delta_3 d_{it-1}^F$		-0.113* (0.0495)	-0.0119 (0.0425)	0.0519 (0.116)
Country fixed effects	✓	✓	✓	✓
Distributed lag in Δy	✓	✓	✓	✓
Test for equality of β_{HH} and β_F , p -value		.008	.004	.535
R^2	0.146	0.281	0.114	0.0324
Observations	695	221	341	120

Notes. Column (1) explores nonlinearity in the relation between credit expansion and growth in the full sample. For both household and firm debt, we estimate separate coefficients for positive and negative changes in debt to GDP. Columns (2)–(4) report separate regressions by de facto exchange rate arrangement in year t from [Iizetzki, Reinhart, and Rogoff \(2010\)](#). Fixed regimes cover arrangements with no separate legal tender, currency boards, pegs, and narrow horizontal bands (coarse ERA code 1 from [Iizetzki, Reinhart, and Rogoff \(2010\)](#)). Intermediate regimes include crawling pegs, crawling bands, moving bands, and managed floats (coarse ERA codes 2 and 3). We exclude 11 country-years in which the de facto arrangement is classified as “freely falling” (cases where 12-month inflation is greater than 40%). All regressions include country fixed effects and three lags of GDP growth. Reported R^2 values are from within-country variation. Standard errors in parentheses are dually clustered on country and year. +, *, ** indicate a significance at the 0.1, 0.05, 0.01 levels, respectively.

TABLE X
UNEMPLOYMENT AND HOUSEHOLD DEBT EXPANSIONS

	Full sample		Fixed ER regimes	Intermediate	Freely floating
	$\Delta_3 u_{it+3}$ (1)	$\Delta_3 u_{it+3}$ (2)	$\Delta_3 u_{it+3}$ (3)	$\Delta_3 u_{it+3}$ (4)	$\Delta_3 u_{it+3}$ (5)
$\Delta_3 d_{it-1}^{HH}$	0.132** (0.0380)	0.105** (0.0390)	0.264** (0.0736)	0.0709+ (0.0367)	-0.0159 (0.0587)
$\Delta_3 d_{it-1}^F$	0.0363* (0.0153)	0.0373* (0.0158)	0.0615+ (0.0319)	0.0385* (0.0194)	0.0403 (0.0332)
Country fixed effects	✓	✓	✓	✓	✓
Distributed lag in Δu		✓	✓	✓	✓
Test for equality of β_{HH} and β_F , p -value	.026	.131	.001	.425	.506
R^2	0.145	0.207	0.397	0.235	0.254
Observations	662	638	211	296	120

Notes. This table reports regression estimates of the change in the unemployment rate from t to $t+3$ on the change in household and nonfinancial firm debt to GDP from $t-4$ to $t-1$. All columns include country fixed effects. Columns (2)–(5) include three lags of the change in the unemployment rate as controls. Columns (3)–(5) estimate the regression across exchange rate regimes, as defined in [Table IX](#). Reported R^2 values are from within-country variation. Standard errors in parentheses are dually clustered on country and year. +, *, ** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

Read:

- Table IX shows nonlinearity: positive household debt growth predicts lower future GDP growth, but negative household debt growth does not symmetrically predict higher growth.
- The coefficient on positive household debt growth is about -0.436 ; the coefficient for nonpositive household debt growth is close to zero.
- The effect is strongest under fixed exchange rates, with coefficient about -0.534 ; it is weaker under intermediate regimes and close to zero under floating regimes.
- Table X shows that household debt booms predict higher future unemployment.
- A one-standard-deviation increase in household debt growth predicts about **0.82 percentage points** higher unemployment.

5.10 Table XI: house prices

Read:

- Household debt booms and real house price growth are strongly related: the coefficient in column (1) is about **1.035**.
- But household debt remains a robust predictor of lower future growth even after controlling for house price growth.

- House prices matter, but the paper argues that household debt is the stronger and more robust predictor in the horse race.
- The interpretation is that credit supply shocks may drive both debt and house prices, rather than house prices alone driving the main result.

TABLE XI
HOUSE PRICES, HOUSEHOLD DEBT, AND GDP GROWTH

	$\Delta_3 \ln(P_{it-1}^{Housing})$	$\Delta_3 y_{it+3}$			
	(1)	(2)	(3)	(4)	(5)
$\Delta_3 d_{it-1}^{HH}$	1.035** (0.263)	-0.243** (0.0658)	-0.241** (0.0709)	-0.210** (0.0481)	-0.179* (0.0742)
$\Delta_3 d_{it-1}^F$		-0.0443 (0.0379)	-0.0575 (0.0381)	-0.0676 (0.0429)	-0.0348 (0.0348)
$\Delta_3 \ln(P_{it-1}^{Housing})$		-0.0607** (0.0224)	-0.0688* (0.0280)	-0.0552+ (0.0333)	-0.0288 (0.0269)
Country fixed effects	✓	✓	✓	✓	✓
Distributed lag in Δy			✓	✓	✓
Year fixed effects					✓
Sample	Full	Full	Full	Pre-2006	Full
R^2	0.109	0.175	0.184	0.184	0.502
Observations	514	514	514	395	514

Notes. This table explores the relationship between household debt and house prices and presents robustness to including house price growth in the main specification. Column (1) shows the correlation between the increase in household debt to GDP and real house price growth over $t-4$ to $t-1$. Real house price growth is constructed from the BIS's "Long series on nominal residential property prices" (fourth quarter value) deflated by the CPI. Columns (2)–(5) report results from robustness checks that include the change in real house prices from $t-4$ to $t-1$ in the main specification. Columns (3)–(5) control for three GDP growth lags. All columns include country fixed effects, and column (5) also includes year fixed effects. Reported R^2 values are from within-country variation. Standard errors in parentheses are dually clustered on country and year. +, *, ** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

TABLE XII CREDIT EXPANSION, NET EXPORTS, AND CORRELATION WITH THE GLOBAL HOUSEHOLD DEBT CYCLE									
	$\Delta_3 \ln \frac{N_{it+3}}{Y_t}$ (1)	$\Delta_3 \ln \frac{X_{it+3}}{Y_t}$ (2)	$\Delta_3 \ln \frac{M_{it+3}}{Y_t}$ (3)	$\Delta_3 \ln \frac{S_{it+3}}{Y_t}$ (4)	$\Delta_3 \ln \frac{MC}{Y_t}$ (5)	$\Delta_3 \ln \frac{N_{it+3}}{Y_t}$ (6)	$\Delta_3 \ln \frac{X_{it+3}}{Y_t}$ (7)	$\Delta_3 \ln \frac{M_{it+3}}{Y_t}$ (8)	$\Delta_3 \ln \frac{S_{it+3}}{Y_t}$ (9)
$\Delta_3 d_{it-1}^{HH}$	0.17** (0.049)	0.39** (0.15)	-0.061 (0.13)	-0.23 (0.16)	-0.076* (0.035)	0.049 (0.052)	0.11* (0.057)	-0.22* (0.090)	0.25** (0.039)
$\Delta_3 d_{it-1}^F$	0.022 (0.021)	0.12* (0.069)	-0.003 (0.057)	-0.005 (0.063)	0.013 (0.015)	0.031* (0.019)	0.023 (0.020)	-0.045 (0.036)	0.024 (0.020)
$\Delta_3 d_{it-1}^{HH} \times \text{openness}_{it}$						0.17** (0.027)	0.14** (0.036)		
$\Delta_3 d_{it-1}^{HH} \times \rho^{Global}$								-0.33 (0.22)	-0.22** (0.071)
Global: $-\Delta_3 d_{it-1}^{HH}$									-0.74** (0.26)
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Distributed lag in Δy		✓	✓	✓	✓	✓	✓	✓	✓
Year fixed effects							✓		
Test for equality of β_{HH} and β_F , p-value	.013	.144	.868	.359	.026				
R^2	0.062	0.075	0.021	0.039	0.056	0.080	0.19	0.16	0.080
Observations	695	695	695	695	695	695	695	693	693

Notes. This table reports regressions of a variety of outcomes from t to $t+3$ on the expansion in household and nonfinancial firm debt to GDP from $t-4$ to $t-1$. The dependent variable in column (1) is the change in net exports from t to $t+3$ relative to GDP in year t . Column (2) uses the change in log exports minus log imports over the same period as the dependent variable. Columns (3) and (4) show results for the change in exports and imports relative to initial GDP. The dependent variable in column (5) is the change in the share of consumption imports in total imports. Columns (6) and (7) interact the change in household debt with a country's openness to international trade, openness, defined as the average imports plus exports to GDP ratio during the sample period. Column (8) focuses on three-year-ahead growth and includes the interaction of the increase in household debt with ρ^{Global} , the correlation between country i three-year household debt expansion and the sample average household debt expansion excluding country i given by equation (5). Column (9) reports the same regressions with the change in net exports from t to $t+3$ relative to GDP in year t as the dependent variable. Column (10) includes the global average change in household debt to GDP over $t-4$ to $t-1$ excluding country i . All regressions include country fixed effects and column (7) includes year fixed effects. Reported R^2 values are from within-country variation. Standard errors in parentheses are clustered at the country level. +, *, ** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

5.11 Table XII, Table XIII, and Figure VII: external adjustment and the global debt cycle

Read:

- Table XII shows that countries improve net exports after a household debt boom, mainly through lower imports rather than higher exports.
- More open economies use the external margin more strongly after household debt booms.
- But countries whose household debt cycles are highly correlated with the global cycle have less ability to export their way out of the downturn.
- Table XIII shows a global household debt cycle: global household debt growth predicts lower global GDP growth.
- The coefficient on global household debt growth is about -1.09 ; a one-standard-deviation rise in global household debt growth predicts roughly **2.2 percent** lower global GDP growth over the next three years.
- Figure VII visualizes this global relation: household debt, not firm debt, is the global predictor of subsequent growth.

TABLE XIII
GLOBAL HOUSEHOLD AND FIRM DEBT AND GLOBAL GROWTH

	Dependent variable: global average Δy_{t+3}				
	(1)	(2)	(3)	(4)	(5)
Global $\Delta_3 d_{t-1}^{HH}$	-1.094** (0.300)		-1.097** (0.311)	-0.966** (0.252)	-0.928** (0.288)
Global $\Delta_3 d_{t-1}^F$		-0.103 (0.192)	0.00896 (0.177)	-0.0756 (0.149)	0.0727 (0.192)
Global Δy_{t-1}				0.341 (0.244)	0.342 (0.257)
Global Δy_{t-2}				0.390+ (0.224)	0.426* (0.189)
Global Δy_{t-3}				0.477+ (0.258)	0.532+ (0.280)
Sample	Full	Full	Full	Full	Pre-2006
Test for equality of β_{HH} and β_F , p -value					
R^2	0.295	0.007	0.295	0.471	0.426
Observations	46	46	46	46	40

Notes. This table reports time series regressions of the sample average real GDP growth from t to $t+3$ on the sample average change in household and firm debt to GDP from $t-4$ to $t-1$. Newey-West standard errors in parentheses are computed with six lags. +, *, ** indicate significance at the 0.1, 0.05, 0.01 levels, respectively.

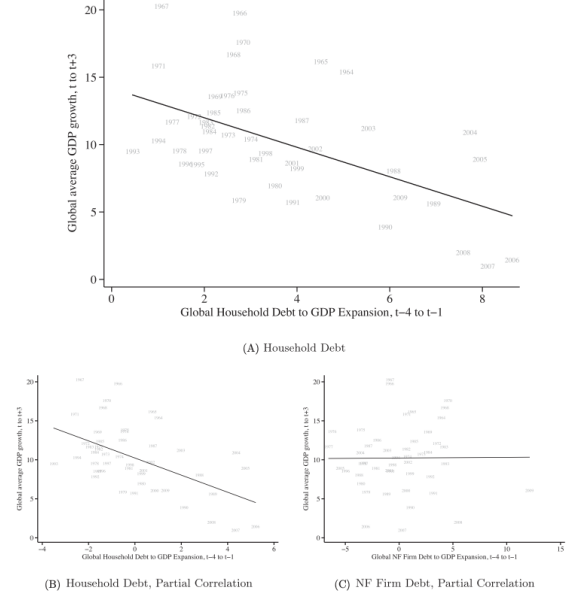


FIGURE VII

Global Household Debt Expansions and Global Growth

This figure illustrates the relationship between the sample average of real GDP growth from t to $t+3$ and the sample average of the change in household and firm debt to GDP from $t-4$ to $t-1$. Each point refers to year t . In Panel B household debt is partialled out with the expansion in nonfinancial firm debt, while in Panel C nonfinancial firm debt is partialled out with the expansion in household debt.

6 Short summary

- **Method.** The paper combines panel VARs, local projections, medium-run predictive regressions, spread-based proxy SVARs, cross-sectional IV episodes, forecast-error regressions, and global-cycle regressions.
- **Core fact.** A three-year increase in household debt to GDP predicts lower GDP growth over the next three years. Firm debt does not generate the same delayed boom-bust pattern.
- **Mechanism.** The evidence is most consistent with credit supply shocks: household debt booms occur when mortgage spreads are low, and spread-induced debt booms predict future slowdowns.
- **Expectations.** IMF and OECD forecasters do not fully internalize the downside risk of household debt booms, so household debt predicts negative forecast errors.
- **Frictions.** The bust is stronger when monetary adjustment is constrained, especially under fixed exchange rates, and the bust shows up in unemployment.
- **Global result.** When household debt booms are synchronized globally, countries cannot all use net exports to offset the downturn, so the global household debt cycle predicts global growth slowdowns.

7 Conclusion

7.1 Core conclusion

- Household debt booms are powerful predictors of future business-cycle weakness across countries.
- The household sector is special: household debt booms generate a consumption-led expansion and a delayed contraction, unlike nonfinancial firm debt.
- The evidence favors a model in which credit supply expands too much, partly because lenders or investors underestimate downside risk.

7.2 Economic intuition

- During the boom, lenders offer cheaper and more abundant mortgage credit.
- Households borrow and consume more, house prices rise, imports rise, and output is temporarily supported.
- When credit conditions reverse or households begin deleveraging, consumption demand falls.
- If wages are sticky or monetary policy cannot adjust enough, the fall in demand becomes lower output and higher unemployment.
- If many countries deleverage at the same time, external adjustment is limited because not everyone can run higher net exports simultaneously.

7.3 Takeaway

This paper turns the household-debt view of the Great Recession into a broad international business-cycle fact. Its central message is that household credit booms are not merely side effects of growth expectations. They often reflect credit supply expansions and optimistic beliefs, and when these booms reverse, macroeconomic frictions turn household deleveraging into lower growth and higher unemployment.